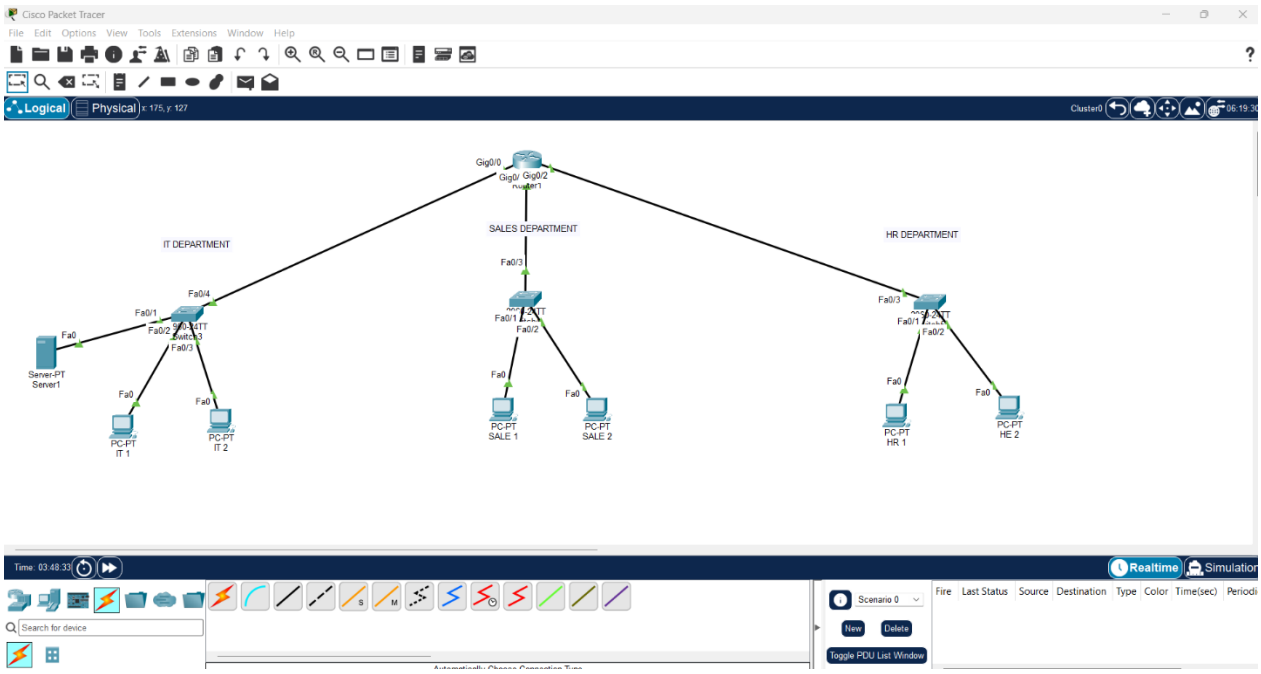
SAMUEL LETSU

CS-C26-09-NA-010

PROJECT WORK 2

TechNova LTD. Network in Cisco Packet Tracer

I set up a full network topology for TechNova LTD. with 1 router (2911) and 3 switches (2960) for 3 departments: IT, Sales, and HR. I included 6 PCs, 2 per department. I added 1 server to the IT department and connected all devices using straight-through cables.

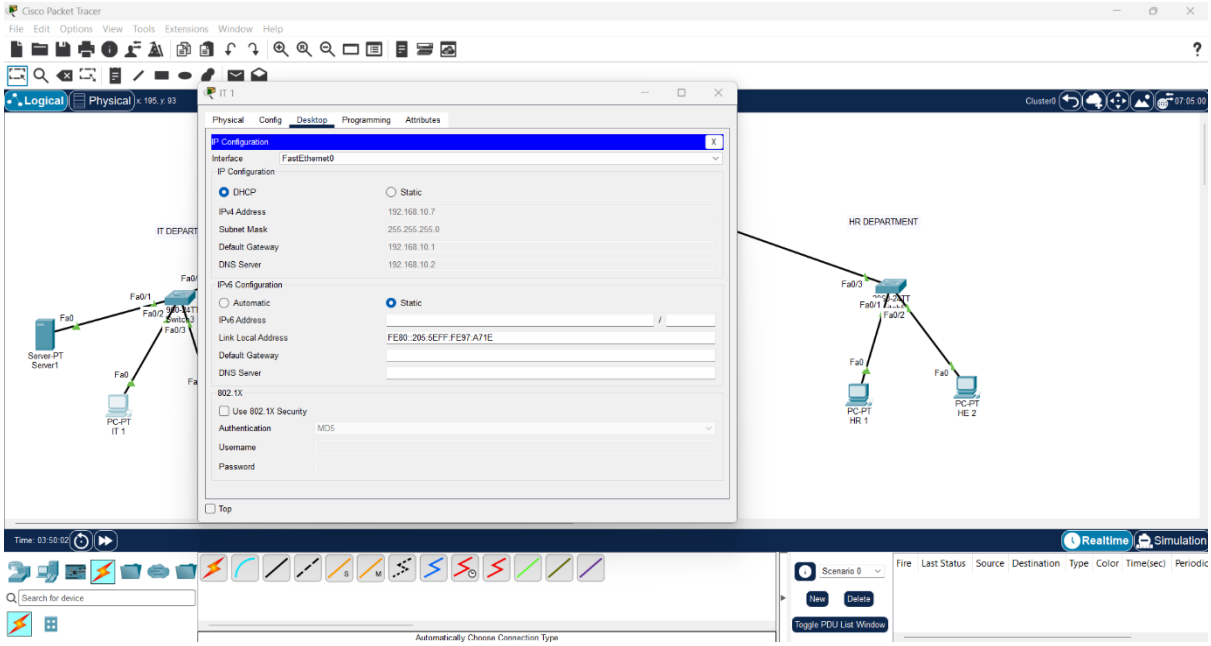


I assigned IP addresses to every device using DHCP and configured all router interfaces:

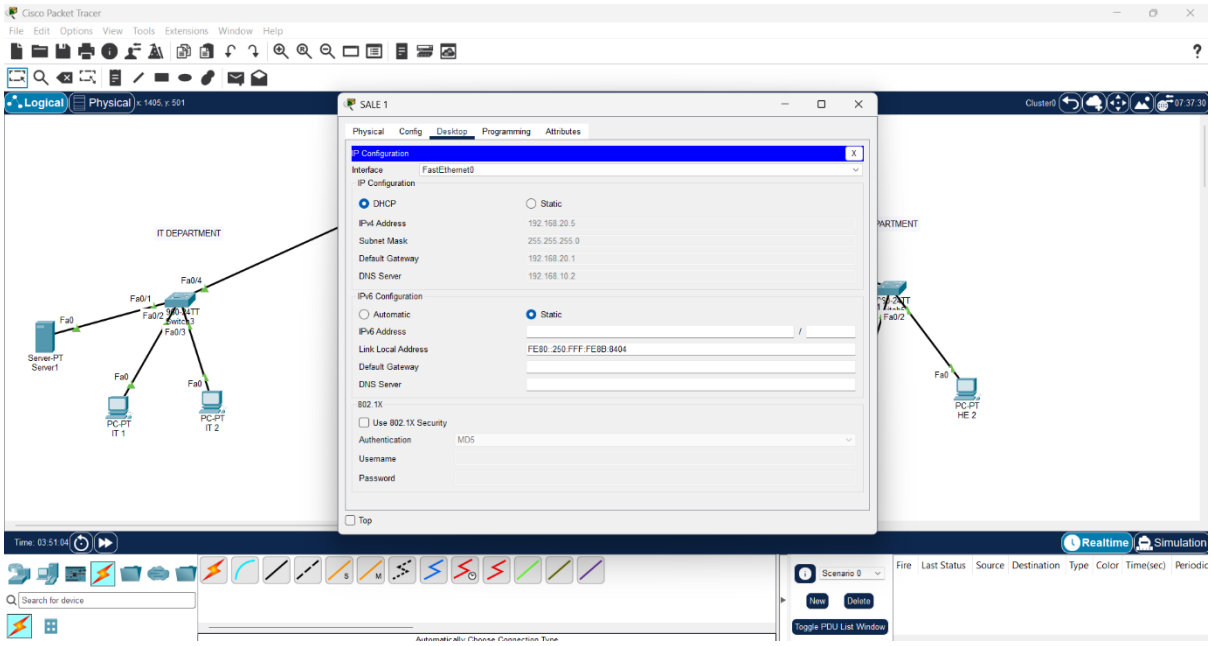
I set the default gateway for each department appropriately and configured a static DNS server.

Below are screenshots of the IP address configuration of the three departments.

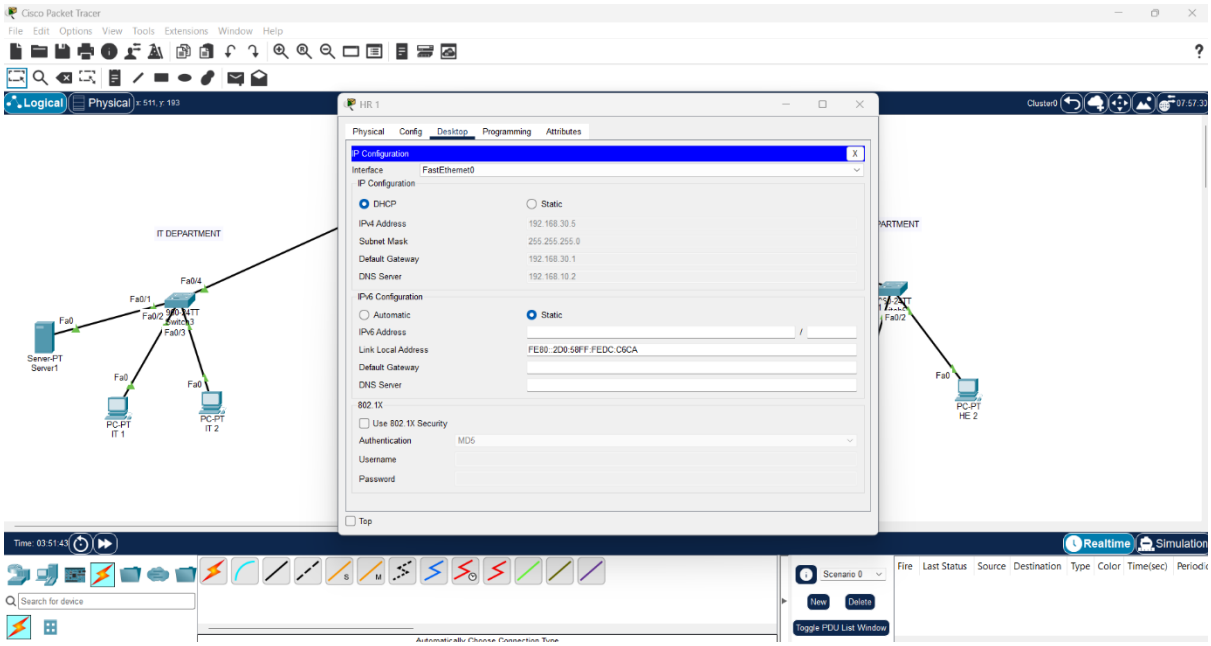
IT Department IP address configuration



Sales Department IP address configuration

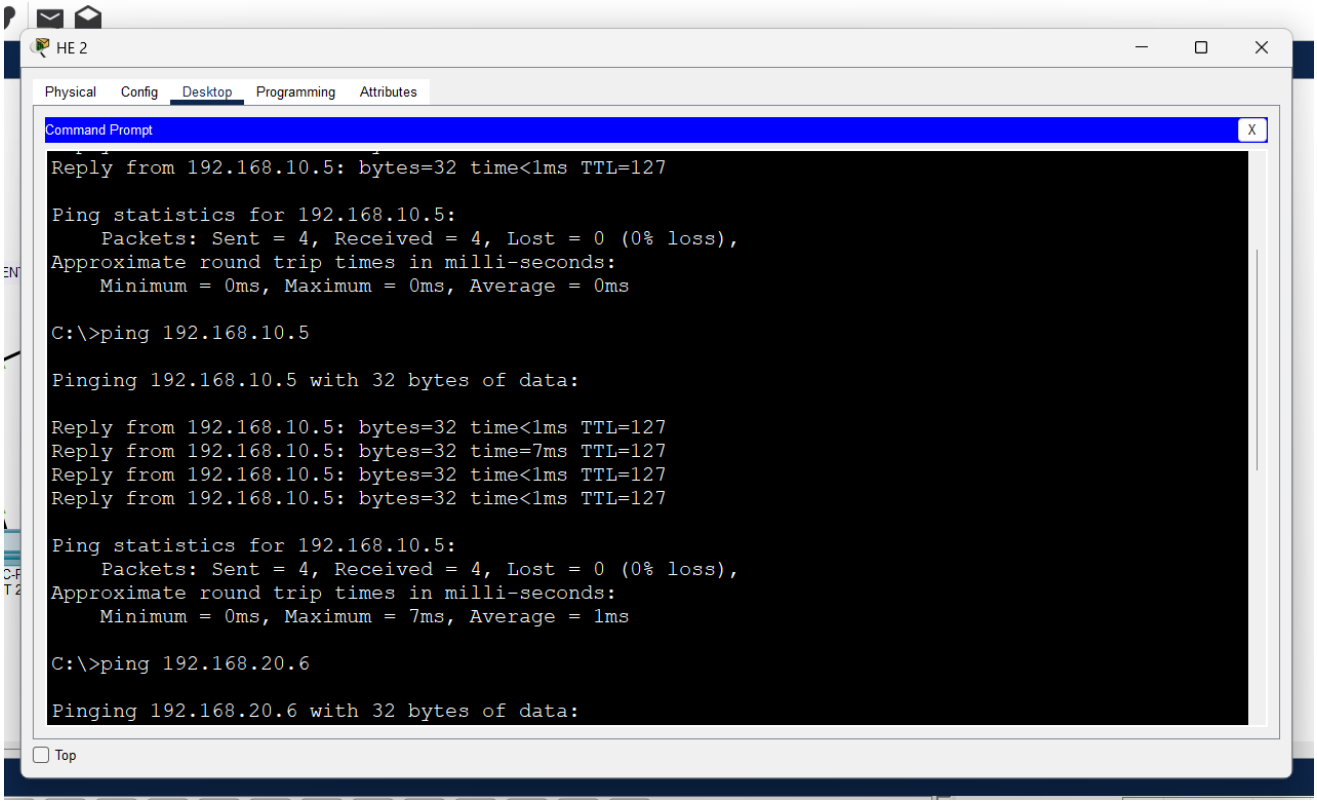


HR Department IP address configuration



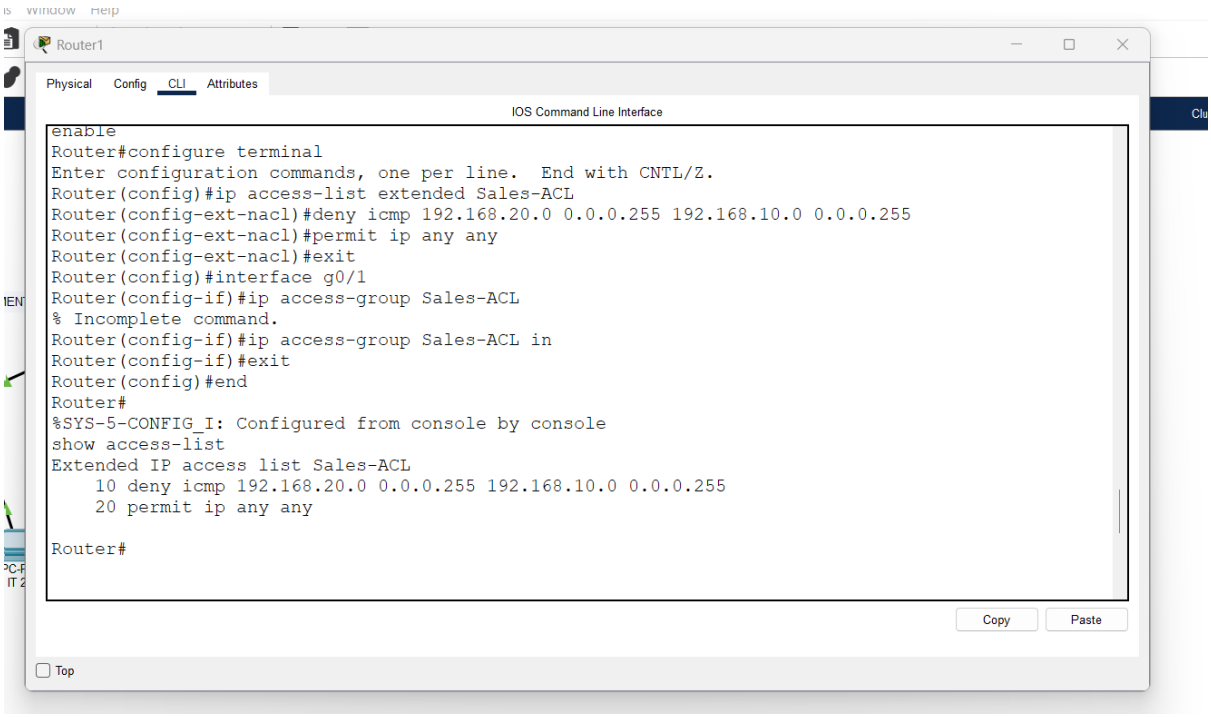
A basic connectivity test was carried out for all three departments using **ping** to confirm that all devices can communicate before applying any restrictions.

Below is a successful ping from the HR PC to the IT PC



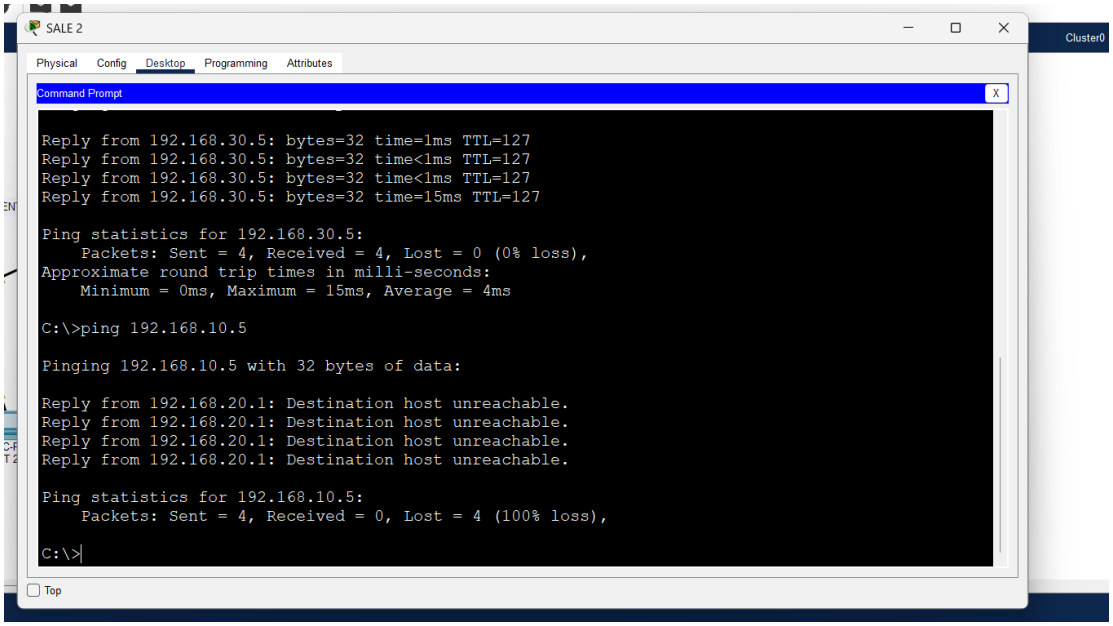
I then applied an ACL command on the router so that the **SALES Department** (1922.168.20.0/24) can't **ping** the **IT Department** (192.168.10.0/24) but will have access to the server services, while all other traffic remains allowed.

Below is the screenshot of the ACL command entered in the router CLI



I ran a ping test to confirm the ACL command is working correctly. I ran a ping test from the Sales Department to the IT department, which resulted in destination host unreachable, meaning our ACL command is working correctly.

Below is a screenshot of the ping test.



Finally, I tested to see if the **Sales Department** have access to other server services such as HTTP.

Below is the result

